

SAENTIS, Switzerland

WMO: 066800

Lat: 47.2494N

Lon: 9.3436E

Elev: 2500

StdP: 74.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 2	(b) -18.6	(c) -16.7	(d) -33.7	(e) 0.2	(f) -4.4	(g) -30.2	(h) 0.3	(i) -4.8	(j) 22.8	(k) -3.5	(l) 20.8	(m) -4.1	(n) 6.2	(o) 270	(p) 0.936

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.4	(c) 14.9	(d) 9.6	(e) 13.5	(f) 8.8	(g) 12.3	(h) 8.1	(i) 10.9	(j) 13.2	(k) 9.9	(l) 12.3	(m) 9.0	(n) 11.2	(o) 2.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	10.3	11.9	8.8	9.6	10.7	8.0	9.0	9.9	38.8	13.3	35.9	12.3	33.6	11.2	15.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.1	16.1	13.8		-20.3	18.3	2.8	4.3	-22.3	21.4	-23.9	23.9	-25.5	26.4	-27.5	29.5	
			DB	-20.9	13.1	2.6	1.1	-22.8	13.9	-24.3	14.5	-25.7	15.1	-27.6	15.9	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-7.0	-7.4	-5.5	-2.9	1.1	4.8	6.6	6.9	3.7	1.5	-3.3	-5.8
	DBStd	6.70	4.32	4.91	4.50	4.14	3.85	4.35	3.86	3.92	4.07	4.42	4.97	4.62
	HDD10.0	3887	527	486	481	387	276	163	117	110	192	263	398	488
	HDD18.3	6895	786	719	739	637	534	406	364	354	440	521	648	747
	CDD10.0	33	0	0	0	0	0	7	11	14	2	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	2	0	0	0	0	0	2	0	0	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
(14) Wind	WSAvg	5.6	7.3	7.0	6.7	4.8	4.6	4.2	4.3	4.2	4.5	5.5	6.5	7.6
(15) Precipitation	PrecAvg	2518	219	189	201	188	199	238	257	267	190	158	188	235
	PrecMax	3559	574	771	673	458	504	531	490	539	449	415	699	779
	PrecMin	1675	6	19	25	12	51	84	51	48	27	13	0	16
	PrecStd	499	141	134	142	103	104	90	105	113	89	98	124	176
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	5.0	4.9	7.0	11.2	16.7	16.4	16.8	13.3	11.4	7.9	4.7
		MCWB	-1.8	-1.4	-1.8	1.9	6.2	8.9	10.1	10.3	6.6	4.3	0.8	-2.1
	2%	DB	2.1	2.9	3.0	5.4	9.4	13.8	14.5	15.0	11.3	9.8	6.0	3.0
		MCWB	-3.2	-3.0	-2.1	1.1	5.3	8.7	9.8	9.6	6.0	3.5	0.1	-3.3
	5%	DB	0.0	1.0	1.6	4.1	8.0	12.2	13.2	13.6	10.1	8.4	4.5	1.5
		MCWB	-3.8	-3.8	-2.5	0.7	4.4	8.2	9.0	8.9	5.3	2.8	-0.6	-3.4
	10%	DB	-1.5	-1.0	0.3	2.6	6.5	10.7	11.8	12.2	9.0	7.0	3.1	0.0
		MCWB	-4.3	-4.3	-2.7	-0.3	3.6	7.5	8.1	8.3	5.3	2.3	-0.8	-3.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.9	0.9	3.7	7.9	11.6	12.2	12.3	9.2	6.4	3.2	0.0
		MCDB	0.9	4.1	2.0	5.0	10.0	13.9	14.7	14.6	10.9	8.6	5.8	1.1
	2%	WB	-1.5	-1.8	-0.3	2.2	6.3	10.1	10.8	10.9	8.0	5.2	1.8	-1.0
		MCDB	0.1	0.1	1.2	4.2	8.3	12.4	13.3	13.3	9.6	7.5	3.8	0.4
	5%	WB	-2.5	-2.9	-1.2	1.2	5.0	9.0	9.8	9.8	7.0	4.3	0.7	-1.9
		MCDB	-0.9	-0.5	0.2	3.3	7.2	11.2	12.0	12.4	9.0	6.7	2.8	-0.3
	10%	WB	-3.8	-3.9	-2.2	0.1	3.9	7.9	8.9	8.9	6.0	3.5	-0.3	-2.7
		MCDB	-2.0	-1.2	-0.5	2.0	5.8	10.1	11.0	11.2	8.0	5.9	2.0	-0.9
(35) Mean Daily Temperature Range	MDBR		4.6	4.6	4.4	4.0	4.0	4.4	4.4	4.4	4.2	4.4	4.4	4.6
	5% DB	MCDBR	5.5	6.7	5.1	4.2	4.9	6.0	5.8	5.6	5.0	5.1	5.0	5.5
		MCWBR	3.8	4.6	3.7	4.0	4.7	5.6	5.2	5.5	5.0	4.1	3.1	3.8
	5% WB	MCDBR	5.9	6.3	4.9	4.0	4.9	5.4	5.3	5.2	4.7	4.6	4.4	5.3
		MCWBR	5.5	5.4	4.4	4.1	4.7	5.5	5.3	5.3	4.9	4.1	3.8	4.8
(40) Clear-Sky Solar Irradiance	taub		0.210	0.225	0.250	0.275	0.287	0.293	0.294	0.289	0.271	0.253	0.231	0.208
	taud		2.496	2.474	2.506	2.522	2.531	2.537	2.521	2.552	2.600	2.638	2.619	2.519
	Ebn at Noon		933	978	992	987	979	970	964	959	956	933	900	898
	Edh at Noon		59	76	88	96	99	99	100	92	80	65	53	51
(44) All-Sky Solar Radiation	RadAvg		1.25	2.12	3.33	4.36	4.94	5.70	5.55	4.84	3.66	2.33	1.33	1.03
	RadStd		0.14	0.28	0.46	0.64	0.50	0.53	0.50	0.48	0.37	0.26	0.17	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	+0.94	N/A	+0.34	N/A	N/A	N/A	N/A
Regional (121 neighbors)	N/A	N/A	N/A	+0.70	+0.42	+0.34	N/A	N/A	+57	+15

Nomenclature: See separate page